Binomial models 1−6, 30 replicates (suni2D standardize = FALSE, tau grid with 0; uniLasso loo = FALSE)

Entries are mean (standard error) across replicates.

Model 1

uniLasso, AR(1) rho = 0.8, class mean shift in x1−x20; n = 100, p = 300, 30 nonzero

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	21.9710 (0.4085)	3.1287 (0.4760)	2.6342 (0.3087)	16.6667 (0.5800)	0.9364 (0.0016)	0.4600 (0.0126)	0.0106 (0.0014)
suni2D_lambda_alpha	24.2116 (1.3371)	3.1310 (0.4757)	2.8195 (0.3693)	16.7333 (0.5851)	0.9367 (0.0016)	0.4622 (0.0127)	0.0106 (0.0014)
suni2D_sum_rate	24.1916 (1.3293)	3.1310 (0.4757)	2.8182 (0.3701)	16.7000 (0.5757)	0.9366 (0.0016)	0.4611 (0.0127)	0.0106 (0.0014)
uniLasso	21.8252 (0.4843)	3.1634 (0.4676)	2.5599 (0.3044)	17.2000 (0.5762)	0.9358 (0.0015)	0.4656 (0.0127)	0.0120 (0.0013)
lasso	22.5458 (0.2326)	3.7854 (0.6869)	2.9383 (0.3318)	29.5000 (0.9764)	0.8988 (0.0029)	0.4856 (0.0136)	0.0553 (0.0031)
adalasso_uni	21.7270 (0.4361)	3.1974 (0.5164)	2.5474 (0.3074)	17.0667 (0.5507)	0.9358 (0.0015)	0.4633 (0.0127)	0.0117 (0.0012)
adalasso_lasso	29.0024 (0.6618)	4.0191 (0.5566)	3.7697 (0.4482)	13.0333 (0.6093)	0.9148 (0.0019)	0.2911 (0.0106)	0.0159 (0.0018)

Model 2

independent; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	5.3779 (0.2339)	0.7387 (0.0576)	0.5425 (0.0288)	27.8000 (2.2547)	0.9796 (0.0020)	0.8700 (0.0236)	0.0193 (0.0021)
suni2D_lambda_alpha	5.3793 (0.2343)	0.7378 (0.0576)	0.5419 (0.0288)	27.9000 (2.2577)	0.9795 (0.0020)	0.8700 (0.0236)	0.0194 (0.0021)
suni2D_sum_rate	5.3805 (0.2342)	0.7383 (0.0576)	0.5421 (0.0288)	27.7667 (2.2533)	0.9796 (0.0020)	0.8700 (0.0236)	0.0193 (0.0021)
uniLasso	4.8047 (0.1997)	0.7358 (0.0417)	0.6912 (0.0396)	63.1667 (1.8650)	0.9461 (0.0019)	0.9633 (0.0102)	0.0541 (0.0019)
lasso	5.8566 (0.1907)	0.7492 (0.0529)	0.5739 (0.0242)	56.0667 (3.7704)	0.9528 (0.0037)	0.9433 (0.0141)	0.0471 (0.0038)
adalasso_uni	4.7987 (0.1994)	0.7358 (0.0417)	0.6901 (0.0396)	63.0333 (1.8616)	0.9462 (0.0018)	0.9633 (0.0102)	0.0539 (0.0019)
adalasso_lasso	6.9154 (0.5099)	0.7127 (0.0468)	0.9526 (0.0732)	31.0000 (1.2512)	0.9773 (0.0012)	0.9167 (0.0167)	0.0221 (0.0012)

Model 3

AR(1) rho = 0.5; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.6031 (0.0823)	1.0761 (0.0444)	0.4848 (0.0160)	4.9000 (1.3142)	0.9874 (0.0010)	0.1133 (0.0229)	0.0038 (0.0012)
suni2D_lambda_alpha	9.6030 (0.0824)	1.0746 (0.0444)	0.4848 (0.0160)	4.8667 (1.3163)	0.9874 (0.0010)	0.1133 (0.0229)	0.0038 (0.0012)
suni2D_sum_rate	9.6029 (0.0824)	1.0754 (0.0445)	0.4847 (0.0160)	4.9333 (1.3156)	0.9873 (0.0010)	0.1133 (0.0229)	0.0038 (0.0012)
uniLasso	9.8817 (0.2506)	1.0087 (0.0408)	0.8372 (0.0533)	55.3667 (2.7137)	0.9424 (0.0026)	0.3867 (0.0248)	0.0520 (0.0026)
lasso	9.6251 (0.0775)	1.0724 (0.0449)	0.4820 (0.0151)	10.9667 (2.3937)	0.9822 (0.0021)	0.1600 (0.0274)	0.0095 (0.0022)
adalasso_uni	9.8529 (0.2366)	1.0120 (0.0406)	0.8313 (0.0501)	55.1333 (2.6530)	0.9425 (0.0025)	0.3833 (0.0245)	0.0518 (0.0026)
adalasso_lasso	10.2865 (0.2475)	1.0651 (0.0408)	0.7861 (0.0726)	8.9667 (1.8178)	0.9840 (0.0015)	0.1467 (0.0261)	0.0076 (0.0017)

Binomial models 1−6, 30 replicates (suni2D standardize = FALSE, tau grid with 0; uniLasso loo = FALSE)

Entries are mean (standard error) across replicates.

Model 4

block, size 20, rho = 0.5; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.0467 (0.2356)	1.1722 (0.0545)	0.6043 (0.0163)	10.1333 (2.6716)	0.9841 (0.0019)	0.2133 (0.0425)	0.0081 (0.0023)
suni2D_lambda_alpha	9.0436 (0.2356)	1.1693 (0.0552)	0.6042 (0.0162)	10.1667 (2.6743)	0.9841 (0.0019)	0.2133 (0.0425)	0.0081 (0.0023)
suni2D_sum_rate	9.0416 (0.2356)	1.1686 (0.0556)	0.6041 (0.0162)	10.1667 (2.6743)	0.9841 (0.0019)	0.2133 (0.0425)	0.0081 (0.0023)
uniLasso	7.4899 (0.2614)	0.9069 (0.0459)	0.7919 (0.0444)	58.0333 (2.3985)	0.9453 (0.0022)	0.6667 (0.0277)	0.0519 (0.0023)
lasso	8.1662 (0.2850)	0.9692 (0.0570)	0.5574 (0.0181)	33.5667 (4.7145)	0.9664 (0.0037)	0.4967 (0.0550)	0.0289 (0.0043)
adalasso_uni	7.4723 (0.2618)	0.9044 (0.0458)	0.7857 (0.0430)	57.9000 (2.3559)	0.9454 (0.0022)	0.6667 (0.0277)	0.0518 (0.0023)
adalasso_lasso	10.9449 (0.8593)	0.9291 (0.0630)	1.0330 (0.0986)	22.0000 (2.5194)	0.9777 (0.0016)	0.4833 (0.0532)	0.0173 (0.0021)

Model 5

imbalanced, AR(1) rho = 0.5, prevalence 0.2; n = 200, p = 1000, 10 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.8090 (0.0759)	0.4096 (0.0303)	0.5089 (0.0194)	4.1667 (1.3494)	0.9877 (0.0010)	0.0933 (0.0235)	0.0033 (0.0012)
suni2D_lambda_alpha	9.8102 (0.0751)	0.4075 (0.0304)	0.5089 (0.0194)	4.2333 (1.3685)	0.9876 (0.0010)	0.0933 (0.0235)	0.0033 (0.0012)
suni2D_sum_rate	9.8094 (0.0752)	0.4075 (0.0304)	0.5088 (0.0194)	4.2000 (1.3505)	0.9877 (0.0010)	0.0933 (0.0235)	0.0033 (0.0012)
uniLasso	11.1883 (0.3221)	0.6602 (0.0489)	1.0105 (0.0760)	48.1333 (2.7412)	0.9477 (0.0026)	0.2900 (0.0251)	0.0457 (0.0027)
lasso	9.7811 (0.0631)	0.3985 (0.0271)	0.5071 (0.0186)	10.2333 (2.8598)	0.9823 (0.0024)	0.1267 (0.0299)	0.0091 (0.0027)
adalasso_uni	11.1818 (0.3147)	0.6553 (0.0490)	1.0093 (0.0737)	49.0000 (2.7127)	0.9468 (0.0026)	0.2900 (0.0251)	0.0466 (0.0027)
adalasso_lasso	12.1084 (0.9787)	0.6014 (0.0544)	0.9271 (0.1159)	7.0000 (1.7604)	0.9854 (0.0014)	0.1200 (0.0277)	0.0059 (0.0016)

Model 6

Breheny−Huang (2011), independent; n = 400, p = 1000, 4 nonzero (+/−1)

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	0.4210 (0.0271)	0.0518 (0.0088)	0.0475 (0.0036)	9.4333 (0.6741)	0.9946 (0.0007)	1.0000 (0.0000)	0.0055 (0.0007)
suni2D_lambda_alpha	0.4200 (0.0270)	0.0528 (0.0090)	0.0475 (0.0036)	9.5000 (0.6708)	0.9945 (0.0007)	1.0000 (0.0000)	0.0055 (0.0007)
suni2D_sum_rate	0.4207 (0.0269)	0.0528 (0.0090)	0.0475 (0.0036)	9.4667 (0.6674)	0.9945 (0.0007)	1.0000 (0.0000)	0.0055 (0.0007)
uniLasso	1.5328 (0.2000)	0.2157 (0.0235)	0.2312 (0.0278)	70.8333 (4.7416)	0.9332 (0.0047)	1.0000 (0.0000)	0.0671 (0.0048)
lasso	0.9486 (0.0310)	0.1064 (0.0135)	0.1052 (0.0050)	39.1000 (4.1410)	0.9649 (0.0041)	1.0000 (0.0000)	0.0352 (0.0042)
adalasso_uni	1.5234 (0.1979)	0.2146 (0.0230)	0.2296 (0.0273)	70.7333 (4.7112)	0.9333 (0.0047)	1.0000 (0.0000)	0.0670 (0.0047)
adalasso_lasso	1.0831 (0.1665)	0.1452 (0.0169)	0.1659 (0.0240)	20.3000 (2.0275)	0.9837 (0.0020)	1.0000 (0.0000)	0.0164 (0.0020)